Md Zahidul Islam

Ph.D. Candidate, Industrial Engineering (Operations Research)
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Education

Clemson University — Clemson, South Carolina, USA
Ph.D. Candidate, Industrial Engineering (Operations Research); GPA: 3.90/4.00 — *Aug 2023 – Expected May 2028*

Advisor: Dr. Kevin M. Taaffe, Harriet and Jerry Dempsey Professor and Chair

University of Tennessee – Knoxville — Knoxville, Tennessee, USA
M.S. in Industrial Engineering; GPA: 4.00/4.00 — *Aug 2021 – Aug 2023*

Rajshahi University of Engineering & Technology (RUET) — Rajshahi, Bangladesh
B.Sc. in Industrial & Production Engineering; CGPA: 3.90/4.00, 1st in class — *Jan 2012 – Nov 2016*

Research Interests

Healthcare operations research and workforce scheduling; large-scale mixed-integer and constraint programming; multi-objective optimization and simulation; healthcare analytics and predictive modeling; human factors and clinician well-being; mixed-methods empirical research.

Awards, Fellowships, and Grants

CECAS GSAB Travel Grant, College of Engineering, Computing and Applied Sciences, Clemson University — 2026
Graduate Student Government Graduate Travel Grant, Clemson University — 2026
INFORMS Student Chapter Annual Award, Magna Cum Laude, as chapter President and Vice President — 2026, 2025
Dr. Lloyd W. Crawford Fellowship, University of Tennessee Space Institute — 2021
University Gold Medal, RUET; highest CGPA in the Department of Industrial & Production Engineering — 2019
Prime Minister Gold Medal, UGC Bangladesh; highest CGPA among all students, College of Mechanical Engineering — 2018

Research Experience

Clemson University — Clemson, SC, USA
Graduate Research Assistant, Department of Industrial Engineering — *Jan 2025 – Present*

- Conduct research on a federally funded, multi-institution project to redesign emergency department clinician shift scheduling to improve well-being and patient outcomes.
- Develop mixed-integer and constraint-programming models for multi-site clinician scheduling, design solution methods for large-scale MILPs, and evaluate feasibility, solution quality, computational performance, and scalability.
- Design clinician surveys, interview guides, and data-collection instruments to elicit scheduling preferences, workload concerns, fatigue experiences, recovery needs, and barriers to well-being.
- Apply qualitative methods, including semi-structured interviewing, inductive thematic analysis, codebook development, interrater reliability assessment, and coding reconciliation.
- Analyze survey and interview data to identify clinician-specific scheduling needs, differences across professional groups, and relationships among schedule characteristics, sleep, fatigue, burnout, and well-being.
- Integrate evidence from optimization models, surveys, interviews, wearable-device data, and physiological measures to support evidence-informed scheduling models and policies.
- Coordinate participant recruitment, survey administration, interviews, wearable-device deployment, physiological sample collection, field logistics, and data tracking with research assistants and clinical collaborators.
- Collaborate with industrial engineers, psychologists, physicians, nurses, and healthcare administrators across multiple institutions to translate clinician perspectives into practical scheduling models and recommendations.

University of Tennessee Space Institute — Tullahoma, TN, USA
Graduate Research Assistant — *Aug 2021 – Jul 2022*

- Developed reinforcement-learning policies for perishable inventory systems with stochastic demand, product deterioration, and quantity discounts.
- Developed an agent-based supply chain simulation in AnyLogic to evaluate information sharing and risk pooling as strategies for mitigating the bullwhip effect.
- Collaborated on a scalable supervised-learning approach for the traveling salesman problem that predicts structural features of optimal solutions and integrates them with exact and heuristic solution methods.
- Collaborated on the formulation of a resource-constrained parallel-machine scheduling MILP using data from a plastics manufacturer, solved with Gurobi.

Publications

Refereed Journal Articles

1. Lyu, Z., **Islam, M. Z.**, & Yu, A. J. (2024). A scalable and adaptable supervised learning approach for solving the traveling salesman problems. *IEEE Transactions on Intelligent Transportation Systems*, 25(11), 17092–17104. https://doi.org/10.1109/TITS.2024.3410691
2. Asadujjaman, M., Hossain, M. G., Rana, M. S., & **Islam, M. Z.** (2021). Stature estimation from handprint measurements: An application to the medicolegal investigation. *Egyptian Journal of Forensic Sciences*, 11(1), 1–13. https://doi.org/10.1186/s41935-020-00215-1
3. Nahar, K., **Islam, M. Z.**, & Islam, M. M. (2016). Cost-effective decisions on service rate at a customer care centre with a multi-server queuing model. *Journal of Modern Science and Technology*, 4(1), 144–153.

Refereed Conference Proceedings

1. **Islam, M. Z.**, Roozeboom, N., Gunderson, P., Li, X., & Yu, A. J. (2022). An agent-based simulation model to mitigate the bullwhip effect via information sharing and risk pooling. In *2022 Winter Simulation Conference (WSC)* (pp. 1636–1647). IEEE. https://doi.org/10.1109/WSC57314.2022.10015288
2. Kabir, M. R., Kamal, M. S., & **Islam, M. Z.** (2021). An improved network design of an open-loop reverse supply chain. In *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)* (pp. 1–6). IEEE. https://doi.org/10.1109/ACMI53878.2021.9528280
3. Rahul, J. K., Chakraborty, S., Khayer, N., Hriday, A. R., Mia, M. L., **Islam, M. Z.**, & Ahmed, T. (2021). Design and re-engineering of an automated solar lawn mower. In *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*. IEEE.
4. **Islam, M. Z.**, Supto, M. R., Asadujjaman, M., & Chakrabortty, R. K. (2021). Sex identification from hand measurements using machine learning. In *2021 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)* (pp. 536–540). IEEE. https://doi.org/10.1109/ICCCIS51004.2021.9397222
5. Fahim, M. R. I., Chyon, F. A., Suman, M. N. H., & **Islam, M. Z.** (2020). Time befitting supply chain for the future of Bangladesh apparel industry by aligning Industry 4.0. In *Proceedings of the International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE20)*, Khulna, Bangladesh.

Manuscripts in Preparation

1. **Islam, M. Z.**, Kerivin, H., & Taaffe, K. M. Multi-hospital emergency department physician scheduling with heterogeneous individual preferences.
2. **Islam, M. Z.**, Meyer, S. D., Vosika, E. C., Hirsh, E. L., Taaffe, K. M., Fowler, L. A., Pirrallo, R. G., Britt, T. W., & Prabhu, V. G. Scheduling preferences and their alignment with actual schedules, fatigue, and burnout among emergency department nurses and attending physicians: A multi-site cross-sectional study.
3. **Islam, M. Z.**, Meyer, S. D., Hirsh, E. L., Tilman, M., Taaffe, K. M., Britt, T. W., Vosika, E. C., Prabhu, V. G., Fowler, L. A., & Pirrallo, R. G. The schedule that fits a clinician's life is not the schedule that fits their body: A comparative qualitative study of emergency nurses and physicians.
4. Abrol, S. V., **Islam, M. Z.**, Prabhu, V. G., & Taaffe, K. M. Large language models for thematic analysis in healthcare research: Evaluating prompting strategies against human reference codebooks.

Conference Presentations and Posters

Accepted and Forthcoming

Islam, M. Z., Taaffe, K. M., & Kerivin, H. (2026, November). *Emergency department physician scheduling with heterogeneous individual preferences* [Accepted oral presentation]. INFORMS Annual Meeting, San Francisco, CA.

Islam, M. Z., Meyer, S. D., Vosika, E. C., Hirsh, E. L., Taaffe, K. M., Fowler, L. A., Pirrallo, R. G., Britt, T. W., & Prabhu, V. G. (2026, October). *Preferences, alignment, fatigue, and burnout: A cross-sectional shift-schedule survey of emergency department nurses and attending physicians* [Accepted poster presentation]. Prisma Health Education and Research Institute Research Showcase.

Norton, C., **Islam, M. Z.**, Meyer, S. D., Vosika, E. C., Milne, I., Britt, T. W., Fowler, L. A., Hirsh, E. L., Taaffe, K. M., & Prabhu, V. G. (2026, October). *Shift preference alignment, workload, and fatigue in emergency medicine clinicians* [Accepted poster presentation]. Prisma Health Education and Research Institute Research Showcase.

Presented

Islam, M. Z., Taaffe, K. M., Prabhu, V. G., Meyer, S. D., Hirsh, E. L., Britt, T. W., Vosika, E. C., Fowler, L. A., & Pirrallo, R. G. (2026, May). *Multi-site physician scheduling incorporating work-related fatigue and individual preferences* [Oral presentation]. IISE Annual Conference & Expo, Arlington, TX.

Islam, M. Z., Taaffe, K. M., Prabhu, V. G., Meyer, S. D., Hirsh, E. L., Britt, T. W., Vosika, E. C., Fowler, L. A., & Pirrallo, R. G. (2026, February). *Fatigue-aware multi-site rostering model for emergency department physicians with fairness and well-being constraints* [Oral presentation]. Healthcare Systems Process Improvement Conference, Atlanta, GA.

Islam, M. Z., Meyer, S. D., Hirsh, E. L., Tilman, M., Taaffe, K. M., Britt, T. W., Vosika, E. C., Prabhu, V. G., Fowler, L. A., & Pirrallo, R. G. (2025, November). *The impact of emergency department shift scheduling on clinician fatigue and well-being: A qualitative study* [Poster presentation]. Prisma Health Education and Research Institute Showcase, Columbia, SC.

Owens, A., **Islam, M. Z.**, & Yu, A. J. (2023, May). *Unrelated parallel machine scheduling problem for plastic injection molding* [Conference presentation]. IISE Annual Conference & Expo, New Orleans, LA.

Islam, M. Z., & Yu, A. J. (2023, May). *Deep reinforcement learning for inventory optimization of perishable products with a quantity discount* [Conference presentation]. IISE Annual Conference & Expo, New Orleans, LA.

Teaching Experience

Clemson University Clemson, SC, USA

Graduate Teaching Assistant, Department of Industrial Engineering *Aug 2023 – Dec 2024*

- Supported Systems Modeling (IE 4820), Quality Engineering (IE 4610), Supply Chain Design and Control (IE 6560), and Engineering Economic Analysis (IE 3840) through grading, laboratory and problem-solving sessions, and student consultation.

University of Tennessee – Knoxville Knoxville, TN, USA

Graduate Teaching Assistant, Department of Industrial and Systems Engineering *Aug 2022 – Jul 2023*

- Supported instruction, grading, and student consultation for Facilities Planning and Material Handling (IE 401) and Quality Control and Management (IE 300).

Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh

Lecturer and Assistant Professor, Department of Industrial & Production Engineering *Feb 2019 – Aug 2021*

- Taught undergraduate courses as instructor of record in Probability and Statistics, Operations Management, Product Design, Control Theory and Automation, and Industrial Simulation, with the associated laboratory courses.
- Advised undergraduate research and capstone projects in optimization, simulation, scheduling, manufacturing, and supply chain systems; developed course materials, examinations, and laboratory exercises.

Ahsanullah University of Science and Technology (AUST) Dhaka, Bangladesh

Lecturer, Department of Mechanical and Production Engineering *Apr 2017 – Dec 2018*

- Taught undergraduate courses as instructor of record in Operations Management, Engineering Economy, Material Handling and Maintenance Management, Manufacturing Processes, Machine Tools and Machining, and Engineering Materials.

Research Mentoring

Clemson University Creative Inquiry Clemson, SC, USA

Undergraduate Research Mentor *Fall 2025 – Present*

- Mentor twelve undergraduate researchers across five disciplines (industrial engineering, biological sciences, bioengineering, health science, and nursing) on study procedures, field data collection, data quality, analysis, and research presentation.

Mentees: Ella M. Cheney (Industrial Engineering, Fall 2025–Spring 2026); Brooklyn Brown (Health Science, Fall 2025–Spring 2026); Alexandra March (Biological Sciences, Fall 2025–present); Mason LaVern Wimer (Industrial Engineering, Fall 2025–Spring 2026); Ranitha D. Kumarasinghe (Bioengineering, Fall 2025–present); Ava Grace Meredith (Industrial Engineering, Spring 2026–present); Emily Jackson (Industrial Engineering, Spring 2026–present); Rachael Neill (Industrial Engineering, Fall 2026–present); Deekshita Vemuri (Biological Sciences, Fall 2026–present); Nathan Michael Elm (Biological Sciences, Fall 2026–present); Christopher Wilkinson (Industrial Engineering, Fall 2026–present); Emma Hughes (Nursing, Fall 2026–present).

Leadership and Professional Service

President, Clemson University INFORMS Student Chapter *Aug 2025 – Sep 2026*

Led chapter programming, invited seminars, professional-development and outreach activities, and coordination with faculty and university partners; co-hosted and produced the *Optimality Gap* podcast.

Vice President, Clemson University INFORMS Student Chapter *Aug 2024 – Jul 2025*

Supported chapter programming, professional-development events, and outreach; co-hosted and produced the *Optimality Gap* podcast.

Journal Reviewer: *Journal of Big Data*; *Scientific Reports*; *BMC Medical Imaging.*

Professional Memberships: INFORMS; IISE *2022 – Present*

RUET: Moderator, RUET IPE Club (2019–2021); Organizing Secretary, Association of Industrial & Production Engineering; Vice President, RUET IPE Club; General Secretary, Greater Khulna Association; Class Representative *2012 – 2021*

Technical and Research Skills

Optimization and Simulation: Mixed-integer linear programming, constraint programming, multi-objective optimization, decomposition and branch and bound, Markov decision processes, discrete-event and agent-based simulation; Gurobi, CPLEX, Google OR-Tools, AMPL, Pyomo, AnyLogic, Arena

Programming and Machine Learning: Python (NumPy, pandas, SciPy, Statsmodels), R, MATLAB, C, SQL, Git, LaTeX; PyTorch, TensorFlow, scikit-learn, XGBoost; Clemson Palmetto HPC cluster (SLURM)

Statistical and Mixed Methods: Regression and mixed-effects models, design of experiments, survey design and analysis, semi-structured interviewing, inductive thematic analysis, interrater reliability; Qualtrics, ATLAS.ti, SPSS, Minitab

Last updated: October 2026